Introduction

Software development is rarely an individual activity. Most real-world applications are developed by teams that require collaboration, effective communication, and structured approaches. In this assignment, we describe how two peer collaborate to design the structure of a C programming project. The project was not only technical but also an exercise in teamwork, decision-making, & problem-solving.

By working together, we learned how to share responsibilities, document design decisions and ensure that project was easy to understand, extend, & maintain.

Objective of the Project.

The main objective of our collaborative project was to design a modular c application with a well-defined structure. We selected a student Management System as our project case study because it provided scope for multiple functionalities, such as adding students, viewing records, and searching data.

Specific goals included:

*) Designing a clean file and modular structure for c programs.

*) practicing header files & modular programming.

*) Improving our teamwork & decision-making skills.

*) Version control and documentation.

Team collaboration process

We followed a structured collaboration process which involved:

Role Assignment

Peer A: Focused on coding main features & logic.

Peer B: Worked on documentation, modular design and testing.

Communication methods:

* Weekly in-person discussions.
* Daily updates using messaging apps.
* Task tracking through a shared spread sheet

Planning stages:

1 Brainstroming project ideas.

2 Selecting a manageable project

3 Designing project structure

4 Coding and integration

5 Testing and documentation.

This approach ensured that each member had ownership while also being accountable for group outcomes.

Decision - Making Approach

We adopted a democratic decision - making process. Whenever there was a disagreement, we discussed alternatives, weighed pros & cons and then finalized by consensus.

Examples:

Project choice : out of three ideas (calculator, Library system, student Management), we chose Student Management system because it was both challenging and achievable.

File Structure :

After considering a single-file approach vs modular design, we agreed that modularity improves scalability.

Tools used :

We used GCC compiler, code:: Blocks IDE, and GitHub for version control.

Project structure

we designed the following structure:

```
Student Management/
|
|- include
|   └──── student.h
|
|
|- src/
|   |── main.c
|   |── student.c
|   └── utils .c
|
|
|── docs/
|   └── project_report.docx
|
|
|── Makefile
└── README.md
```

Explanation:

include/ → Header files containing function prototypes and structure.

src/ → source files implementing functionalities.

docs/ → Documentation for project.

Makefile → Automates compilation.

README.md → provides overview for new users.

sample code snippets

Student.h

```
#ifndef STUDENT_H
#define STUDENT_H

typedef struct
{ int id;
  char name[50];
  float marks;
  students;
}

void addstudent (student s);
void display students ();
student searchstudent (int id);
```

```
#endif
```

student.c

```
# include <Stdio.h>
# include "student.h"

# define MAX 100
student students [MAX];
int count =0;

void addstudent( student s)
{
  student [count++] = s;
}

void display students ()
{ for (int i=0; i<count; i++)
  { printf("ID: %d, Name: %S, Marks; %.2f\n,"
    students [i].id,
```

```
Students[i].name, Students[i].marks);
}
}
```

main.c

```
# include <Stdio.h>
# include "Student.h"

int main()
{
    student S1 = {1, "Alice", 85.5};
    student S2 = {2, "Bob", 90.0};

addstudent(S1);
addstudent(S2);

printf("Student Records:\n");
display Students();
return 0;
}
```

challenges faced and solutions.

*) Conflict in design choices - We resolved it through structured discussions.

*) Code integration Issues - Solved by testing each module separately before merging.

*) Time Manggement - Used a shared calendar and divided task fairly.

*) Version Control Conflicts - Learned how to resolve git merge conflicts.

Conclusion & Reflection.

We created a modular C project with a professional file structure, while also practicing teamwork skills such as take division, conflict resolution, and time management.